

Muckund Sharma

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EDUCATION

University of Waterloo

Waterloo, ON

Bachelor of Applied Science, Mechanical Engineering

Sep. 2025 – Apr. 2030

cGPA: 3.98/4.0 • President's Scholarship of Distinction (\$5,000), Mechanical Engineering Leadership Award (\$1,500)

Relevant Coursework: Mechanics, Material Structure and Properties, Control of Material Properties

TECHNICAL SKILLS

CAD, Simulation & Analysis: SolidWorks (CSWA Certified), AutoCAD, CATIA 3DEXPERIENCE (Mechanical Designer Certified), ANSYS Mechanical/Fluent, XFLR5, Xfoil, OpenRocket, FEA, CFD, GD&T, DFM/DFA, Tolerance Analysis, DXF Generation

Manufacturing & Fabrication: CNC Machining (Milling, Lathe), CNC Waterjet, Manual Mill & Lathe, Laser Cutting, 3D Printing (FDM), Composite Layup (Vacuum Infusion), Soldering

Programming & Electronics: Python, C++, MATLAB (Onramp Certified), Arduino, RC Systems, Brushless Motors/ESCs, Servos, Jira

Languages: English (Fluent), French (Intermediate, DELF B1 Certified)

EXPERIENCE

Mechanical Engineer

Sep. 2026 – Present

Waterloo Aerial Robotics Group (WARG)

Waterloo, ON

- Engineered the Pegasus 1 landing gear in SolidWorks (127.8 g per leg) from 22 mm OD carbon-fiber tubes and a 3D-printed PETG-CF bracket, adding an intentional hourglass breakaway neck so the sacrificial joint fails before the airframe or tubes in a hard crash.
- Validated the bracket with orthotropic linear-static FEA in ANSYS (1–4g sweep): 3.81 FOS at 1g and tensile-failure onset at the breakaway neck at 4g, with peak stress changing only 1.22% on mesh refinement.
- Designed a 3D-printable ESC enclosure (APD 120F3) with M3 mounts, component clearances, passive ventilation, and terminal protection.
- Conducted a wingtip-device trade study for the team's 2.42 m, AR 8.5 UAV wing: modeled induced drag against added profile drag for five devices and found each changes wing drag by under 5% (0.17 N at most); NASA Langley wind-tunnel data showed the raked tip with the lowest minimum drag (6.5% below baseline).
- Designed the recommended raked tip in SolidWorks (131 mm long, 200 to 90 mm chord, 54° sweep, 0–10° dihedral; accepted by the team) and defined its XFLR5 VLM validation plan.

Mechanical Engineer

Sep. 2025 – Aug. 2026

Waterloo Rocketry

Waterloo, ON

- Contributed airframe materials testing to Polaris, which set the amateur liquid-fuel altitude world record (63,497 ft, Launch Canada 2026): compression-tested six fiberglass laminates to ASTM D695 / DIN EN 2850 on a 15 kN-capacity system, where the Airstone epoxy laminate came out on top and was adopted for the airframe.
- Developed a Python pipeline that converts load/displacement data into stress-strain curves and failure properties for repeatable comparison across all six laminates.
- Machined a Boeing BSS 7260 compression-testing fixture on a manual mill and lathe and manufactured precision aluminum alignment tabs on a CNC waterjet, holding tight tolerances so coupons align consistently.
- Executed a full vacuum-infusion composite layup end to end, from fabric sequencing and bag assembly to leak testing and resin-flow monitoring.

Project Coordinator & Teaching Assistant

May 2026 – Aug. 2026

York Region Educational Services (YRES)

Markham, ON

- Mentored and evaluated 6 summer program staff through daily check-ins and weekly one-on-one feedback, and built a Jira workspace for a 200+ participant volunteer event with staged tasks and assigned owners.

PROJECTS

MX-01 RC Trainer: Design, Aerodynamic Analysis and Flight Test

Aug. 2026 – Present

- Designed a Cessna 172-inspired RC trainer solo from requirements to flight: SolidWorks CAD and engineering drawing sheet, then a hand-built EPS-foam airframe (0.79 m span, 0.6 kg, 0.12 m² wing) for about \$140; selected, wired, and commissioned the propulsion and control system.
- Sized the aircraft for a 10 m/s cruise at 0.6 kg (Re about 100k, required C_L about 0.80) and selected a Clark Y airfoil from Xfoil polars at Re = 50k, 100k, and 150k, identifying a laminar separation bubble at Re = 50k that cuts peak C_L/C_D from about 53 to about 29.
- Modeled the 3D wing in XFLR5 with viscous polars: peak L/D about 15.0 at 4° (C_L about 0.57) and L/D about 14 at the 10 m/s cruise point; set a 2.3° wing mounting incidence to keep the fuselage near level in cruise.
- Flight-tested over 12 flights (about 1 hr airtime), reducing motor torque-ripple vibration with a second hardboard motor mount and softening over-sensitive controls with 30% transmitter expo.